Left-Handed Generation from Edge-Face Decomposition of the Tetrahedral Channel Algebra

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Abstract

We show that the tetrahedral channel algebra, previously restricted to the six-dimensional edge space C^6, extends naturally to an eight-dimensional space $C^8 = C^6 \oplus C^2$ by incorporating the unique S_4-invariant face-sum direction. Under SU(3) × SU(2), the representation content is (3,2) ⊕ (1,2), matching exactly one left-handed Standard Model generation: a quark doublet in three colors plus a lepton doublet. The lepton sector arises as the kernel of the simplicial boundary operator, providing a topological explanation for color singlet structure. All prior results on the edge space (mass splitting, gluon protection, gauge algebra) are preserved by orthogonality. The face-sum transport eigenvalue lambda = 3/4 yields a distinct lepton mass scale.

1. Introduction

The tetrahedral channel algebra on C^6 has been shown to produce the Standard Model gauge group SU(3) × SU(2) × U(1) as the commutant of the transport matrix T, a mass-splitting mechanism via the swap involution S, and gluon masslessness from eigenspace degeneracy [THM-528–561]. However, C^6 accommodates only the (3,2) representation—a single quark doublet—and lacks color singlets necessary for the lepton sector.

In this work we show that the tetrahedron's face structure provides the missing sector. The four-face space C^4 decomposes under S_4 as C^1(trivial) ⊕ C^3(standard). The trivial component—the face-sum—is the unique color singlet. Tensored with the spin degree of

freedom, it yields C^2 = (1,2), the lepton doublet.

2. Tetrahedral Decomposition

2.1 Edge space

The six edges of the tetrahedron define the channel Hilbert space C^6 with orthonormal basis $\{|\text{sigma}_0\rangle, \ldots, |\text{sigma}_5\rangle\}$. Three orthogonal dimension pairs (opposite edges) partition these into dim 1: $\{\text{sigma}_0, \text{sigma}_5\}$, dim 2: $\{\text{sigma}_1, \text{sigma}_4\}$, dim 3: $\{\text{sigma}_2, \text{sigma}_3\}$.

2.2 Face space

The four faces define a second space C^4 with basis $\{|f_0\rangle, |f_1\rangle, |f_2\rangle, |f_3\rangle\}$, where f_k is the face opposite vertex k.

2.3 Boundary operator

The simplicial boundary operator partial: $C^4 \to C^6$ maps each face to the signed sum of its boundary edges. Its rank is 3 and its kernel is one-dimensional:

$$\ker(\text{partial}) = \text{span}(|f_0\rangle + |f_1\rangle + |f_2\rangle + |f_3\rangle)$$

The image im(partial) is a three-dimensional subspace of C^6 carrying the standard representation of S_4.

3. Face-Singlet Theorem

Definition. Let $|\text{Sigma}\rangle = (|f_0\rangle + |f_1\rangle + |f_2\rangle + |f_3\rangle)/2$.

Proposition. $|\text{Sigma}\rangle$ is the unique direction in C^4 fixed by all elements of S_4.

Proof. The symmetric group S_4 permutes the four faces. The sum of all four basis vectors is invariant under every permutation. Uniqueness follows from Schur's lemma: the trivial

representation of S_4 appears exactly once in $C^4 = C^1(\text{trivial}) \oplus C^3(\text{standard})$. ■

Corollary (Color singlet). Since S_4 permutations of dimensions correspond to color rotations, |Sigma⟩ transforms trivially under $SU(3)_{color}$. It carries no color quantum number.

Boundary vanishing. partial|Sigma⟩ = 0. The face-sum has no boundary in edge space, meaning the color singlet direction has no component in the quark (edge) sector. It is invisible to the color force.

4. Left-Handed Generation Candidate

4.1 Construction

Define $C^8 = C^6 \oplus C^2$, where C^6 is the edge space and C^2 = span{|Sigma, +⟩, |Sigma, –⟩} is the face-singlet tensored with the two spin orientations.

4.2 Representation content

Subspace	Origin	Dim	Representation
C■	6 edges = 3 dim-pairs × 2 spins	6	(3, 2)
C^2	face-singlet × 2 spins	2	(1, 2)
C■	total	8	(3,2) ⊕ (1,2)

Table 1. Representation content of C^8 under $SU(3)_{color} \times SU(2)_{weak}$.

This matches exactly one generation of left-handed Standard Model fermions: a quark doublet (u_L, d_L) in three colors (6 states) plus a lepton doublet (nu_L, e_L) (2 states).

4.3 SU(2) structure on the lepton sector

The swap operator S acts on C^2 by exchanging the two spin states: S|Sigma,+⟩ = |Sigma,–⟩. This is the same SU(2) weak isospin that acts on the quark doublet in C^6. The lepton doublet inherits its weak interaction from the shared spin degree of freedom.

5. Boundary-Operator Bridge

The face space decomposes as C^4 = ker(partial) ⊕ im(partial^T). The boundary operator connects the two sectors:

- **Quarks** = boundary of faces = edges. They carry color because edges connect different faces.
- **Leptons** = the face-sum, which has zero boundary. They are colorless because the total face has no edge-boundary.
- **Leptoquark transitions** correspond to the boundary map partial mixing ker(partial) with im(partial).

This is a precise topological statement: the quark–lepton distinction is the kernel–image decomposition of the simplicial boundary operator on the tetrahedron.

6. Extended Transport and Mass Scales

The transport on C^8 is block-diagonal: $T^8 = T^6 \oplus T_2$. On the IFS, each of four children inherits three of the four parent faces (the fourth becomes the interface). The face-sum transport eigenvalue is therefore $\text{lambda}_{\text{lepton}} = 3/4$.

Sector	T eigenvalue	Mass gap (1−λ)
Vacuum / photon	1	0
Weak (W/Z)	1/2	1/2
Lepton	3/4	1/4
Color (gluons)	1/3	2/3 (degenerate → no mass)

Table 2. Transport eigenvalues and mass gaps by sector.

7. Proof-State Classification

7.1 Exact structural

The following results depend only on the tetrahedral axioms (A1–A3), standard representation theory, and linear algebra. They involve no approximations, free parameters, or phenomenological input:

(i) $C^4 = C^1$(trivial) $\oplus$ C^3(standard) under S_4;(ii) $|\text{Sigma}\rangle$ is the unique S_4-invariant direction (Schur's lemma);(iii) partial$|\text{Sigma}\rangle = 0$, rank(partial) = 3, dim ker(partial) = 1;(iv) $C^8 = C^6 \oplus C^2$ has content (3,2) $\oplus$ (1,2);(v) $\text{lambda}_{\text{lepton}} = 3/4$.

7.2 Supported bridge

The identification of C^6 with the quark doublet and C^2 with the lepton doublet is a bridge claim connecting the algebraic decomposition to observed particle content. The physical interpretation of partial as the quark–lepton connection and the reading of lambda = 3/4 as a lepton mass scale carry interpretive weight.

7.3 Open

Right-handed fermions require $C^{16} = C^8_L \oplus C^8_R$ with SU(2) breaking on the R copy. Yukawa couplings, full anomaly cancellation, and Higgs mass correction remain open. Three-generation structure from dihedral stability applied to C^8 is uncomputed.

8. Forward Extension

The full Standard Model generation requires 16 states. C^8 provides the 8 left-handed states. The remaining 8 right-handed states require a chiral doubling: $C^{16} = C^8_L \oplus C^8_R$. The chirality label derives from the Hodge star signs: L uses (+1,–1,+1), R uses (–1,+1,–1). In the R sector, the SU(2) structure must break so that right-handed fermions are weak singlets. Anomaly cancellation ($\text{Tr } Y^3 = 0$) requires both sectors.

Theorem registry: THM-562. Depends: THM-528 (channels), THM-561 (mass split). Lock status: proof-state footer hard-locked. No downstream render may upgrade bridge claims to exact structural.